$$f\left(\frac{\text{ciclos}}{s}, \frac{\text{vueltas}}{s}\right)$$

$$f\left(\frac{1}{s}\right) = f\left(s^{-1}\right)$$

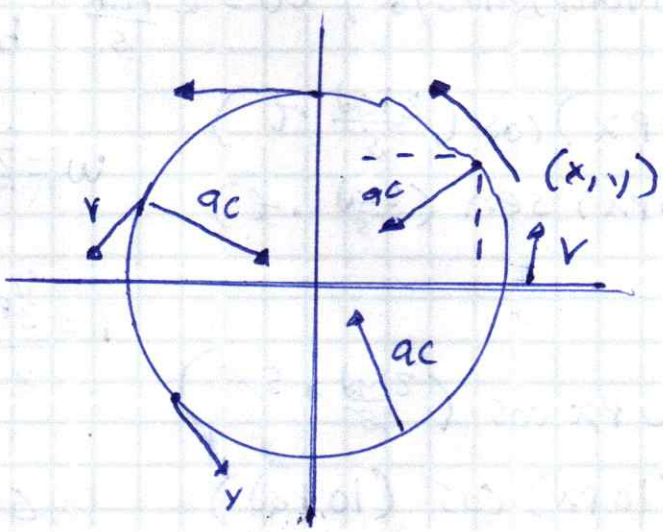
$$s = r\theta$$

$$v = \omega r$$

$$a_c = \frac{v^2}{r} = \omega^2 r$$

$$\omega = \frac{\Delta\theta}{\Delta t}$$

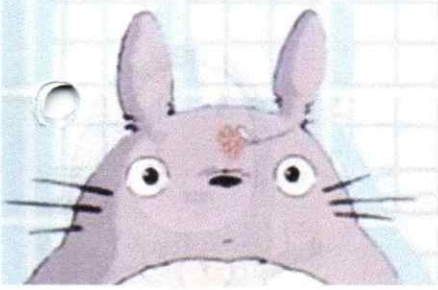
$$\omega = 2\pi f = \frac{2\pi}{T}$$



$$T = \frac{1}{f}, \quad f = \frac{1}{T}$$

$$x(t) = r\cos(\omega t + \varphi)$$

$$y(t) = r\,\text{sen}(\omega t + \varphi)$$



$r = 10\,px$

$6\ FPS = 60$ imagenes/s , $\omega = 2\,\frac{rad}{s}$

$x(t) = (10\,px)\cos\left(\frac{2\,rad}{s}\cdot t\right)$

$y(t) = (10\,px)\,sen\left(\frac{2\,rad}{s}\cdot t\right)$

a) $t = 5s$

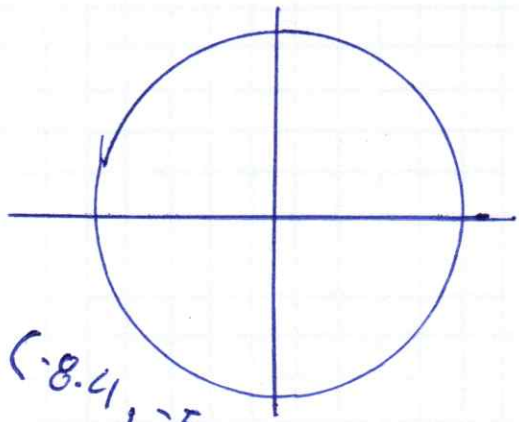
$x(5s) = 10\,px\cos\left(\frac{2\,rad}{\cancel{s}}\times 5\cancel{s}\right)$

$x(5s) = (10\,px)\cos(10\,rad)$

$x(5s) = -8.39\ px = -8.4\ px$

$y(5s) = (10\,px)\,sen(10\,rad)$

$y(5s) = -5.4\ px$



$(-8.4, -5.4)$

b) $60\ \frac{cuadros}{s}\ \times$

$\omega = \frac{2\pi}{T}$, $T = \frac{2\pi}{\omega}$

$T = \frac{2\pi\,rad}{2\,\frac{rad}{s}} = \pi\ seg = \frac{3.14165\,s}{1\ vuelta}$

$60\ \frac{cuadros}{s}\times\frac{3.14165}{vuelta}$

$= 188.5\ \frac{cuadros}{vuelta}$

$V_0 = 90 \frac{km}{h}$, $V_F = 50 \frac{km}{s}$

$r = 150 m$

$a = \frac{\Delta V}{\Delta t} = \frac{V_F - V_0}{t}$

$V_0 = \frac{90.000 m}{3600 s} = 25 m/s$

$V_F = \frac{50.000 m}{3600 s} = 13.9 m/s$

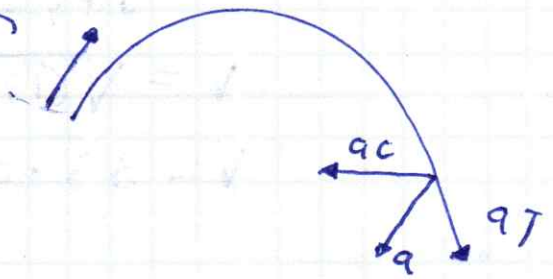
$a = \frac{V_F - V_0}{t} = \frac{13.9 \frac{m}{s} - 25 \frac{m}{s}}{15 s} = -0.74 \frac{m}{s^2}$

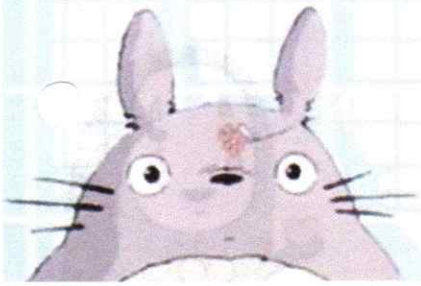
$a_C = \frac{V^2}{R} = \frac{(13.9 \frac{m}{s})^2}{150 m} = 1.28 \frac{m}{s^2}$

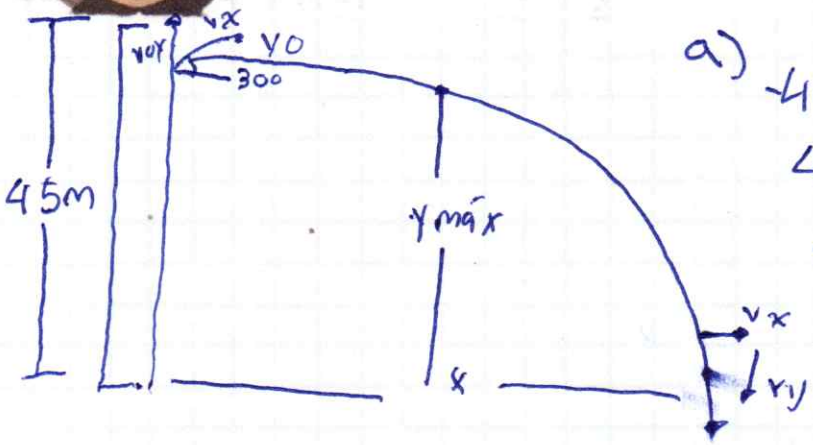
$a_T = -0.74 \frac{m}{s^2}$ tangencial

$a = \sqrt{a_C^2 + a_T^2} = \sqrt{(1.28)^2 + (-0.74)^2}$

$a = 1.478 \frac{m}{s^2} \simeq 1.5 \frac{m}{s^2}$







a) $-45 = 10t - 4.9t^2$

$4.9t^2 - 10t - 45 = 0$

$t = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$= \dfrac{-(10) \pm \sqrt{(-10)^2 - 4(4.9)(-45)}}{2(4.9)}$

$t = \dfrac{10 \pm 31.34}{9.8}$

$t = 4.22$ seg

$x(t) = 17.32t$

$y(t) = 10t - 4.9t^2$

$VFY = 10t - 9.8t^2$

$V^2FY = 100 - 19.6 y_F$

b) $V = \pm\sqrt{V^2x + VFy}$

$VFy = 10 - 9.8(4.22)$

$VFy = -31.36 \frac{m}{s}$

$V = \sqrt{(17.32)^2 + (-31.36)^2}$

$V = 35.8 \, m/s$

$v_0 = 90 \frac{km}{h}$, $v_F = 50 \frac{km}{s}$

$r = 150 m$

$a = \frac{\Delta v}{\Delta t} = \frac{v_F - v_0}{t}$

$v_0 = \frac{90.000 m}{3600 s} = 25 m/s$

$v_F = \frac{50.000 m}{3600 s} = 13.9 m/s$

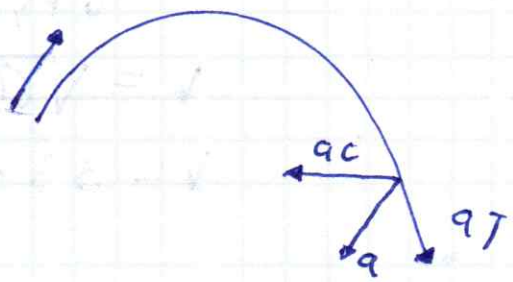
$a = \frac{v_F - v_0}{t} = \frac{13.9 \frac{m}{s} - 25 \frac{m}{s}}{15 s} = -0.74 \frac{m}{s^2}$

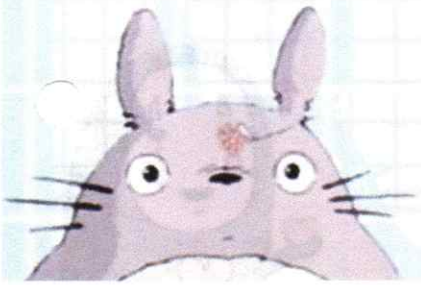
$a_C = \frac{v^2}{R} = \frac{(13.9 \frac{m}{s})^2}{150 m} = 1.28 \frac{m}{s^2}$

$a_T = -0.74 \frac{m}{s^2}$ tangencial

$a = \sqrt{a_C^2 + a_T^2} = \sqrt{(1.28)^2 + (-0.74)^2}$

$a = 1.478 \frac{m}{s^2} \simeq 1.5 \frac{m}{s^2}$





$x(t) = v_x t + x_0$ $\qquad$ $x(t) = 20t$

$y(t) = y_0 + v_{0y} t - 4.9t^2$ $\qquad$ $y(t) = -4.9t^2$

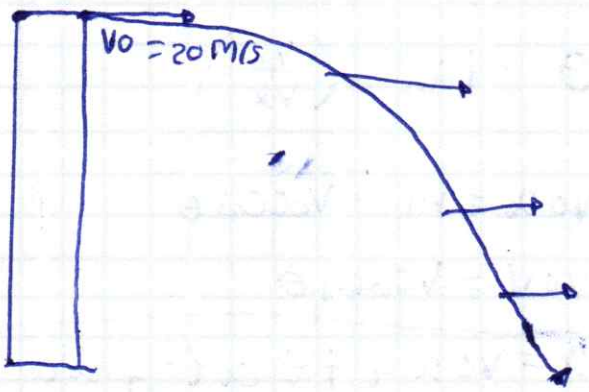
$v_{fy} = v_{0y} - 9.8t$ $\qquad$ $v_{fy} = -9.8t$

$v_{f^2y} = v_{0y}^2 - 19.6(y_f - y_0)$ $\qquad$ $v_{f^2y} = -19.6 y_f$

a) $-45 = -4.9t^2$ $\qquad$ $t^2 = 9.18s$, $t = 3.03s$

b) $v = \sqrt{v_x^2 + v_y^2}$ $\qquad$ $v_{fy} = -9.8(3.03) = -29.7 \frac{m}{s}$

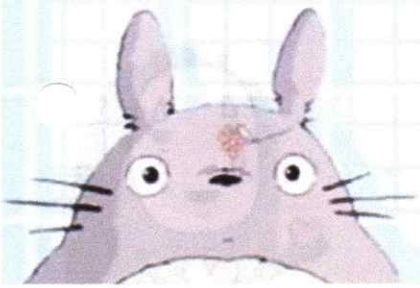
$v = \sqrt{(20)^2 + (-29.7)^2} \approx 35.8 m/s$



c) No es pertinente

d) 45m desde el piso

e) $x_{máx} = 20(3.03) = 60.6m$



$x = v_x t + x_0$

$y = v_{oy} t - \frac{1}{2} g t^2 + y_0$

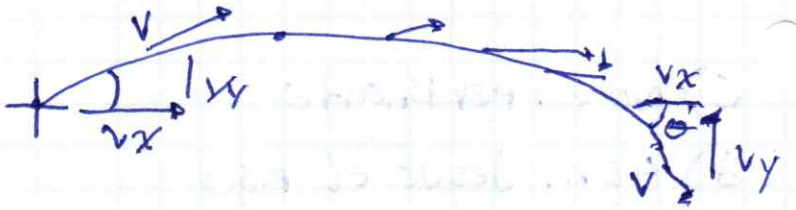
$v_{fy} = v_{oy} - gt$

$v_f^2 y = v_{oy}^2 - 2g(y_f - y_0)$

$v = \sqrt{x^2 + v_y^2}$

$\tan\theta = \frac{v_y}{v_x}$

$\theta = \tan^{-1}\left(\frac{v_y}{v_x}\right)$



$v_{ox} = v_x = v_o \cos\theta$

$v_{oy} = v_o \operatorname{sen}\theta$

$x = v_{ox} t + \frac{1}{2} a t^2 + x_0$

$v_{fy} = v_{ox} + at$

$v_f^2 x = v_{o}^2 x + 2a(x_f - x_0)$

$y = v_{oy} t + \frac{1}{2} a_y t^2 + y_0$

$v_{fy} = v_{oy} + a^2 t$

$v_f^2 y = v_{oy}^2 + 2a(y_f - y_0)$

$v = \sqrt{v_x^2 + v_y^2}$

$\tan\theta = \frac{v_y}{v_x}$

$\theta = \tan^{-1}\left(\frac{v_y}{v_x}\right)$

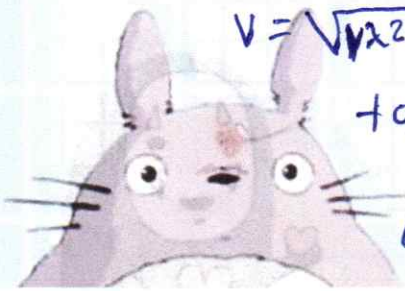
$\vec{v_0} = (6.2, 2.0)\ m/s$, $t = 15s$

$\vec{r_0} = (2.0, 3.0)\ m$ a constante

$v_{15} = (15.0, -1.0)\ m/s$ a calculo de a

b) Halle la dirección aceleracion eye x positivo

c) Determine posición y dirección a los 20s



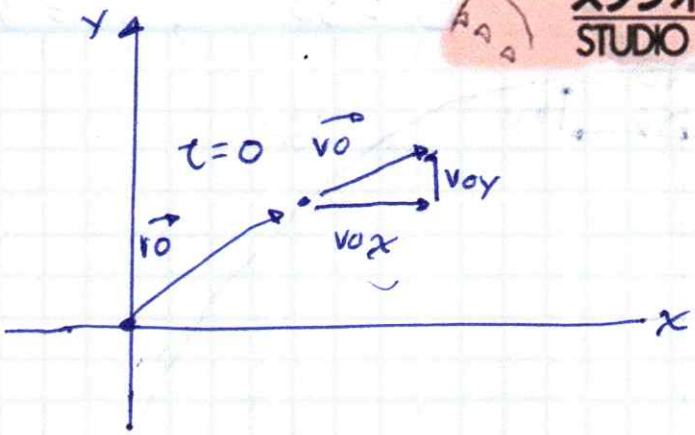
$x = 6.2t + \frac{1}{2} a_x t^2 + 2$

$V_{Fx} = 6.2 + a_x t$

$V_F^2{}_x = 38.41 + a_x t$

$y = 2t + \frac{1}{2} a_y t^2 + 3$

$V_F^2{}_y = 4 + 2a_y (y_F - 3)$



a) $a_x = \frac{V_{Fx} - v_{0x}}{t} = \frac{15 - 6.2}{15} = 0.587 \frac{m}{s^2}$

$a_y = \frac{V_F - v_{0x}}{t} \quad \frac{-1.0 - 2.0}{15} = -0.2 \frac{m}{s^2}$

c) $\vec{r}(t) = \langle x(t), y(t) \rangle$

$\vec{r}(20) = \langle 261.2, 3 \rangle m$

$\theta = \tan^{-1}\left(\frac{y}{x}\right)$

$\theta = 0.658°$

$x(t) = 6.2t \frac{1}{2}(0.587)t^2 + 2.0$

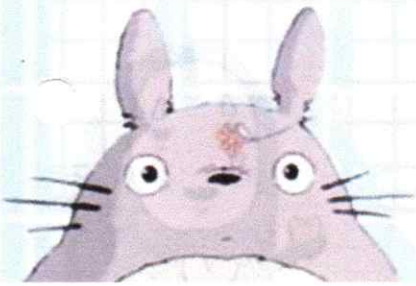
$x(t) = 6.2t + 0.293t^2 + 2.0$

$y(t) = 2.0t + \frac{1}{2}(-0.2)t^2 + 3.0$

$y(t) = 2.0t - 0.1t^2 + 3.0$

$x(20) = 261.2 m$

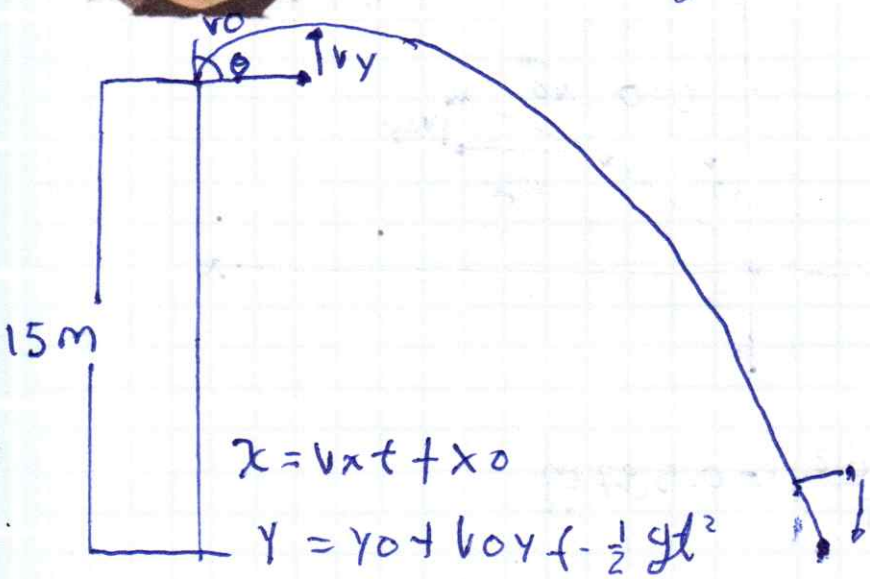
$y(20) = 3 m$



$\theta = 33°$, $V_0 = 30 \frac{m}{s}$

a) 13.62 m

b) 34.55 $\frac{m}{s}$



$x = V_x t + x_0$

$y = y_0 + V_{0y} t - \frac{1}{2} g t^2$

$V_{fy} = V_{0y} - gt$

$V_f^2 y = V_{0y}^2 - 2g(y_f - y_0)$

$\theta = \tan^{-1}\left(\frac{V_y}{V_x}\right)$

$V = \sqrt{V_x^2 + V_y^2}$

$V_x = V_0 \cos\theta$

$V_{0y} = V_0 \operatorname{sen}\theta$

$V_x = 30 \cos 33 = 25.16 \frac{m}{s}$

$V_{0y} = 30 \operatorname{sen} 33 = 16.34 \frac{m}{s}$

$x = 25.16t$

$y = 16.34t - 4.9t^2$

$V_{fy} = 16.34 - 9.8t$

$V_f^2 y = 267 - 19.6 y_f$

a) $0^2 = 267 - 19.6 y_f$

$-267 = -19.6 y_f$

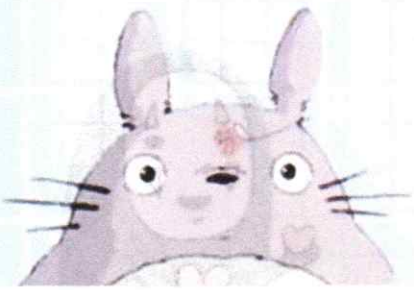
$\frac{-267}{-19.6} = y_f$, $y_f = 13.62 m$

b) $V_{fy}^2 = 561 \frac{m^2}{s^2}$, $V_{fy} = \pm\sqrt{561 \frac{m^2}{s^2}}$

$V_{fy} = -23.68 \frac{m}{s}$

$V = \sqrt{25.16^2 + 23.68^2}$

$V = 34.55 \frac{m}{s}$

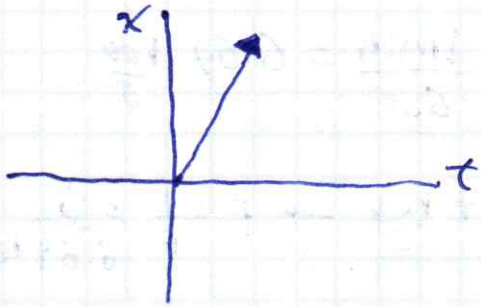


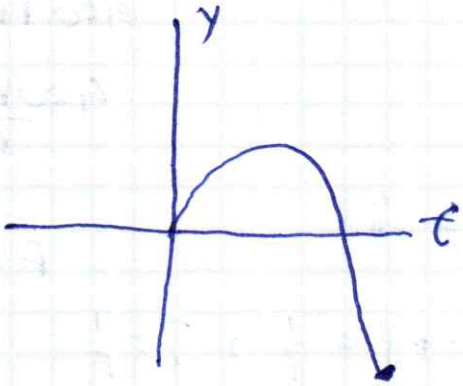
$-23.68 = 16.34 - 9.8t$

$\frac{-23.68 - 16.34}{-9.8} = t, \quad t = 4.085$

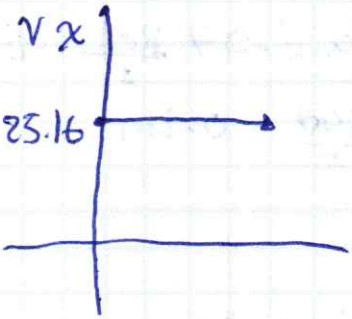
$x = 25.16(4.08) = 102.65m$

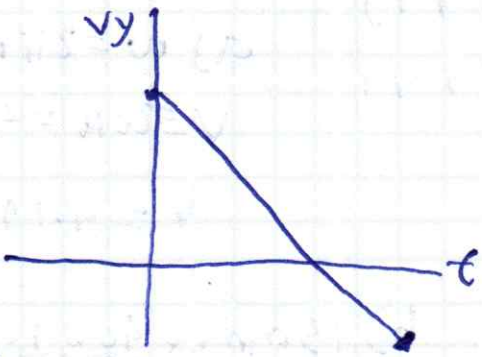
a)

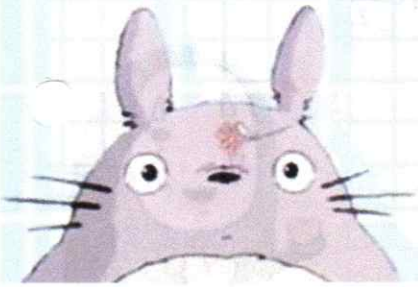




b)







MCU

$S = R\theta$

$V = \omega R$

$a_c = \frac{v^2}{R} = \omega^2 R$

$\omega = \frac{\Delta\theta}{t}$

$\omega = 2\pi f = 2\pi$

$fT = 1 \rightarrow f = \frac{1}{T}, \; T = \frac{1}{f}$

$x(t) = R\cos(\omega t + \varphi)$

$y(t) = R\,\text{sen}(\omega t + \varphi)$

$\theta = \omega t + \theta_0$

$L = 200mm, \; h = 200mm, \; \text{ancho} = 25mm$

$\omega = 41.91 \frac{Rad}{s}, \; V = 4.19 \frac{m}{s}$

400 RPM

$400 \frac{rev}{min} \times \frac{1min}{60s} = 6.67 \frac{rev}{s}$

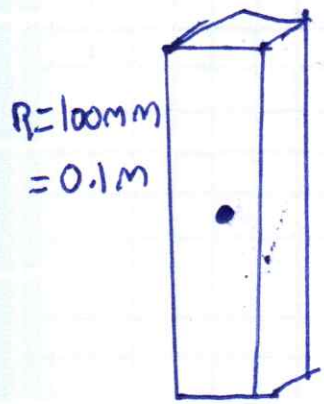
$F = 6.67 \frac{rev}{s} \rightarrow F = \frac{1}{F} = \frac{1s}{6.67 rev} \simeq 0.15s$

$6.67 \frac{\cancel{rev}}{s} \times \frac{2\pi(0.1m)}{1\cancel{rev}} = 4.19 \frac{m}{s}$

a) $\omega = 2\pi f = 2\pi \times 6.67 \frac{Rev}{s} = 41.91 \frac{Rad}{s}$

$V = \omega R = 41.91 \frac{Rad}{s} \times 0.1m$

$V = 4.19 \frac{m}{s}$



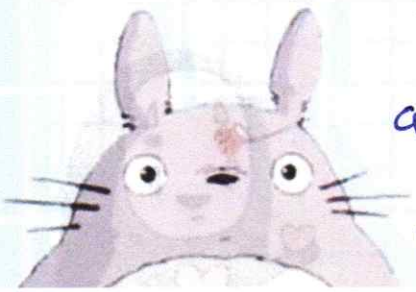
b) $15mm \times \frac{400 rev}{1min} = 600 rev$

$a_{c1} = \frac{v_1^2}{R} = \frac{(4.19 \frac{m}{s})^2}{0.1m} = 175.56 \frac{m}{s^2}$

$a_{c2} = \frac{v_2^2}{R} = \frac{(2 \times 4.19)^2}{0.1m} = 70.224 \frac{m}{s^2}$

$a_{c2} - a_{c1} = 702.24 - 175.56$
$= 526.68 \; m/s^2$

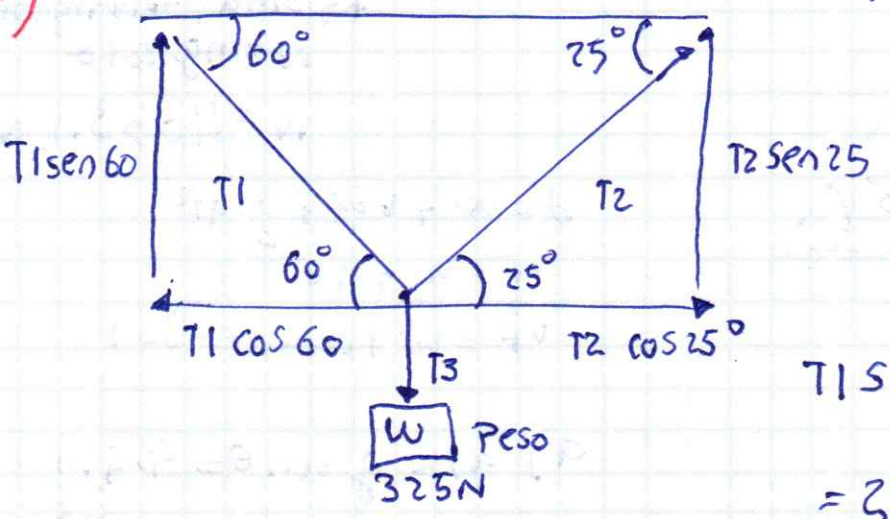
$P = \frac{526.78}{175.56} \times 100\% = 300\%$



$$VF = VO + \frac{P}{100} \times VO$$

$$\left(\frac{VF - VO}{VO}\right) 100\% = P$$

7)



$$\sum Fx = 0 \qquad \sum Fy = 0$$

$$T2\cos 25 - T1\cos 60 = 0 \quad (1)$$

$$T1\,\text{sen}60 + T2\,\text{sen}\,25 - 325N = 0 \quad (2)$$

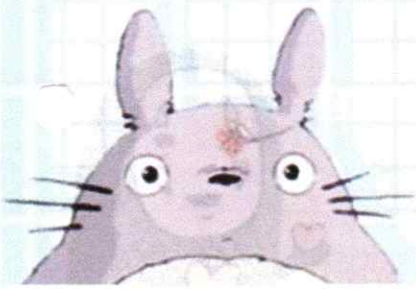
$$T1\,\text{sen}60 + \left(\frac{T1\cos 60}{\cos 25}\right)\text{sen}25 = 325N$$

$$T1\,\text{sen}60 + \left(\frac{T1\cos 60}{\cos 25}\right)\text{sen}$$

$$T1(1.099) = 325N$$

$$T1 = 295.67N$$

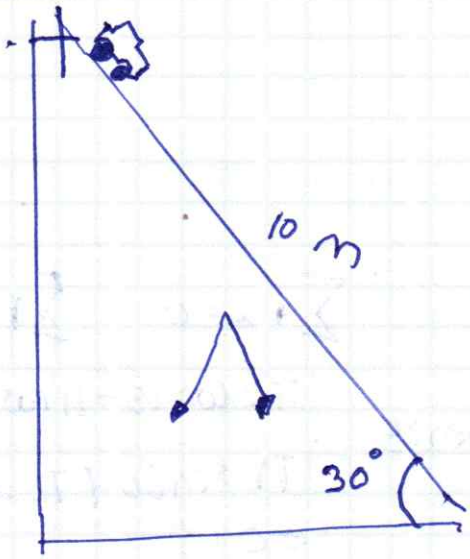
$$T2 = 163.12N$$

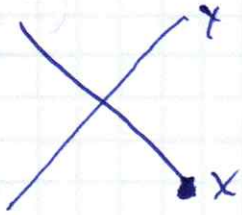


$\sum F = m \cdot a$

8) $m = 1300\,kg$, no tiene fricion

$a = ?$





$f_y = ma_y \rightarrow N - mg\cos\theta = 0$

$N = mg\cos\theta$

$N = 11033.16N$

$d = d_0 + v_0 t + \frac{1}{2} a t^2$

$v_F = v_0 + at$

$v_F^2 = v_0^2 + 2a(d_F - d_0)$

a) $a_y = g \operatorname{sen}\theta = 4.9 \frac{m}{s^2}$

$\sum f_x = ma_x \rightarrow N - mg\cos\theta = 0$

$N = mg\cos\theta$

$\sum f_y = ma_y \rightarrow mg \operatorname{sen}\theta = ma_y$

b) $v_0 = 0$, $d = 10m$

$d = \cancel{v_0 t} + \frac{1}{2} a t^2 + \cancel{d_0}$

$10 = \frac{1}{2}(4.9)t^2$

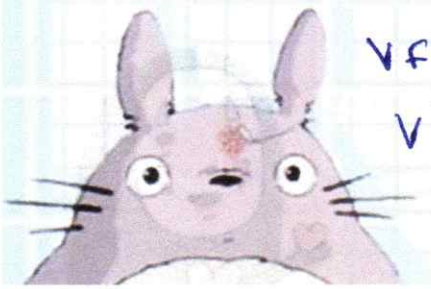
$\frac{2 \times 10}{4.9} = t^2$

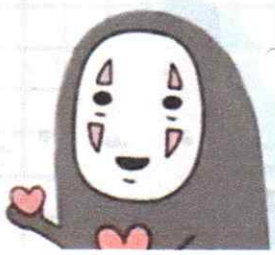
$t = 2.02\,seg$

$v_F = \cancel{v_0} + at$

$v_F = 4.9 \times 2.02$

$v = 9.9 \frac{m}{s}$





movimiento Parabolico

EL movimiento horizontal (x) es rectilineo uniforme (velocidad constante)

el movimiento vertical (y) es un tiro vertical afectado por la gravedad

Descomposición de la velocidad inicial

si lanzamos un objecto con velocidad inicial v_0 formando un ángulo θ:

$v_{0x} = v_0 \cdot \cos(\theta)$ (completamente horizontal)

2 Ecuaciones del movimiento

En x (horizontal):

- posición

$x(t) = v_{0x} \cdot t$

$v_x = v_{0x}$ (constante, no cambia

En y (vertical

$y(t) = v_{0y} \cdot t - \frac{1}{2} g t^2$

$v_y(t) = v_{0y} - gt$

- donde $g = 9.8\ m/s^2$ Hacia abajo

3 fórmulas importantes del movimiento parabólico

1 Tiempo total de vuelo (cuando vuelve al suelo, $y=0$):

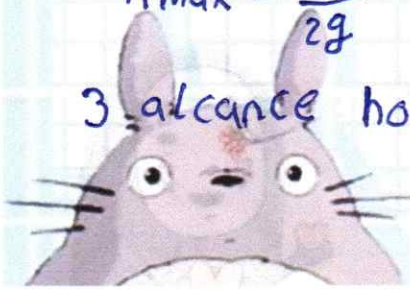
$$T = \frac{2 v_{0y}}{g}$$

2 altura maxima (cuando $v_y = 0$):

$$H_{max} = \frac{v_{0y}^2}{2g}$$

3 alcance horizontal (qué tan lejos llega):

$$R = v_{0x} \cdot T = \frac{v_0^2 \sin(2\theta)}{g}$$



5) $m = 0.6\,kg$, $F = 4N$

$F + F_{riel} = 0$

$F_{riel} = -F$

$= -4N$

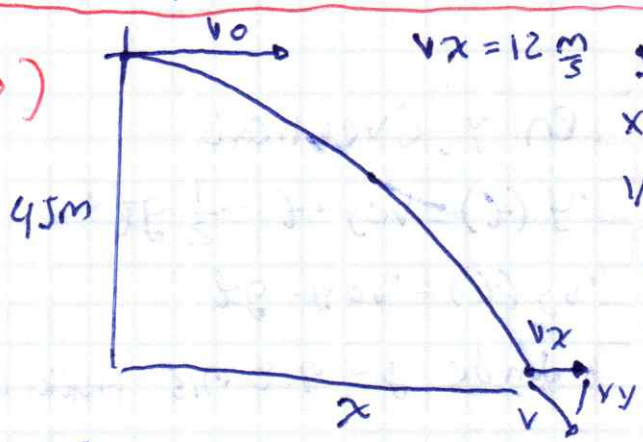
$F = ma$

$F = ma$

$a = \frac{F}{m} = \frac{4N}{0.6\,kg} = \frac{6.67\ kg\frac{m}{s^2}}{kg}$

$= 6.67\frac{m}{s^2}$

6)



$v_x = 12\frac{m}{s}$, $g = 9.81\frac{m}{s^2}$, $t = ?$ $x = ?$ $\vec{v} = ?$

$x = \cancel{x_0} + v_x t \rightarrow x = 12t$

$y = \cancel{y_0} + v_{0y} t - \frac{1}{2} g t^2 \rightarrow y = -4.9t^2$

$v_{Fy} = \cancel{v_{0y}} - gt \rightarrow v_{Fy} = -9.8t$

$v_{Fy}^2 = \cancel{v_{0y}^2} - 2g(y_F - \cancel{y_0}) \rightarrow v_{Fy}^2 = -2g\,y_F$

$\vec{v} = v_x i + v_y j$

$|\vec{v}| = v = \sqrt{v_x^2 + v_y^2}$

$v_{Fy} = -9.8 \times 3.03$

$= -29.7\frac{m}{s}$

$\theta = \tan^{-1}\left(\frac{-29.7}{12}\right)$

$-45 = -4.9t^2$

$\frac{-45}{-4.9} = t^2$, $t = \sqrt{\frac{45}{4.9}} = t = 3.03s$

$x = 12(3.03) = 36.36m$

$\vec{v} = \left(\frac{12m}{s}\right) i - \left(29.7\frac{m}{s}\right) j$

$v = |\vec{v}| = \sqrt{12^2 + (-29.7)^2} = 32.03\frac{m}{s}$

4 Forma de la trayectoria

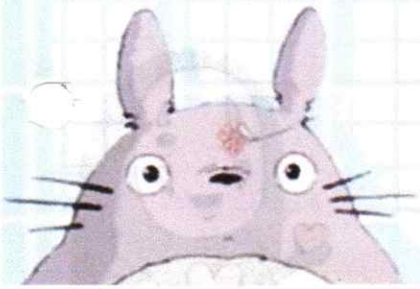
si eliminas t de las ecuaciones, obtienes:

$$y(x) = x \cdot \tan(\theta) - \frac{g}{2 v_0^2 \cos^2(\theta)} x^2$$

es una parábola, por eso movimiento parabolico

5 Pasos para resolver ejercicios

1 identificar v_0 y el angulo θ.

2 calcular v_{0x} y v_{0y}

- Para posiciones en cierto tiempo t.
- Para T. Hmax, R si preguntan por el vuelo completo